Supplementary Information: Segmentation-informed Reconstruction of Crystallization Kinetics and Spatial Fingerprints

This Supplementary Information (SI) provides detailed algorithmic workflows and robustness analyses supporting the main manuscript. In particular, it documents: (i) video-measured kinetics extraction (including burst-window diagnostics), (ii) the morphology-transported kinetics bridge used to connect a single kinetics video to large static segmentation datasets, and (iii) spatially correlated (radially resolved) microstructural fingerprints, including the updated radial kinetic-heterogeneity proxy and the new crowding/impingement descriptors.

1 Input data, file organization, and segmentation format

1.1 Datasets

We analyze two compositions: pristine $FAPbI_3$ (FAPI) and FAPI–TEMPO. Two data streams are used:

1. **Time-resolved video dataset (one per composition).** Bright-field micrographs acquired during FIRA are segmented frame-by-frame to yield time-resolved grain masks. These data are used to extract the video-measured kinetics ($n(t)$, dn/dt, $X(t)$, dX/dt) and burst metrics.

2. **Large static datasets (many micrographs/tiles per composition).** End-state segmentation outputs (grain masks, defect masks, heatmaps) are used to extract morphology distributions and spatial fingerprints with statistical support. These are linked to the single-video kinetics through the morphology-transported kinetics bridge.

1.2 JSON mask format and categories

Instance-level grain, nucleus, and defect annotations are exported as JSON files in a COCO-inspired format.[1, 2] In the present dataset, each JSON is commonly stored as a *list* of detected objects (rather than a full COCO dictionary). Each object contains:

- `segmentation`: run-length encoding (RLE) of a binary mask,
- `bbox`: bounding box $[x_{\min}, y_{\min}, w, h]$,
- `category_id`: class label,
- `score`: confidence score.

Unless otherwise specified, we use:

- crystals (grain body): `category_id = 1`,
- nuclei: `category_id = 2`,

- defects: `category_id = 3`.

All JSON files within a composition folder are treated as one dataset unless micrograph grouping is explicitly used.

2 Mask decoding and per-grain morphology

For each object labeled as a crystal (category 1), the RLE is decoded into a binary mask $M(x, y)$ using `pycocotools`. The following geometry is extracted:

$$A = \sum_{x,y} \mathbf{1}_{M(x,y)}, \tag{1}$$

$$(c_x, c_y) = \frac{1}{A} \sum_{x,y} (x, y)\, \mathbf{1}_{M(x,y)}. \tag{2}$$

Perimeter P is estimated from a one-pixel morphological boundary; circularity is:

$$C = \frac{4\pi A}{P^2}. \tag{3}$$

Defect masks are accumulated into an image-level union mask $M_{\text{defect}}(x, y)$; the defect area within each grain is:

$$A_{\text{def}} = \sum_{x,y} \mathbf{1}_{M \wedge M_{\text{defect}}}, \tag{4}$$

yielding the defect fraction:

$$\phi = \frac{A_{\text{def}}}{A}. \tag{5}$$

3 Video-measured kinetics extraction

3.1 Video-derived observables

From time-resolved masks, we compute:

- event count proxy $n(t)$ (e.g., number of "kept"/active grains at time t),
- event-rate proxy dn/dt,
- transformed fraction proxy $X(t)$ from segmented areas,
- transformation rate dX/dt.

The operational definitions (including filtering/keeping rules, normalization by field-of-view area, and smoothing choices) follow the implementation used to generate `counts_shifted.csv` and the burst-metric extraction script used in the main text.

3.2 Burst-metric extraction and QC

To robustly quantify the dominant crystallization burst from a single video per composition, we compute burst metrics (peak time, duration, FWHM, synchrony metrics, and rate ratios) from the video-derived $X(t)$ and $n(t)$ traces. For numerical stability, we disable running-max monotonic enforcement and floor time at 0 ms (see command-line settings in the project log).

SI_Fig_S1_segmentation_workflow_placeholder.pdf

Figure 1: (Placeholder) SI schematic of the segmentation, curation, and feature extraction workflow. This figure should summarize mask decoding, grain geometry extraction, defect overlap, heatmap sampling, and export of per-grain tables.

SI_Fig_S2_video_Xt_QC_placeholder.png

Figure 2: (Placeholder) Video-measured $X(t)$ quality-control plots for FAPI and FAPI–TEMPO, including any monotonic-envelope checks and normalization choices. Suggested inputs: `qc_X_mono_FAPI.png` and `qc_X_mono_FAPI_TEMPO.png`.

SI_Fig_S3_video_dXdt_dndt_placeholder.png

Figure 3: (Placeholder) Video-measured rate curves and burst annotations (dX/dt and dn/dt) for FAPI and FAPI–TEMPO, including peak-time markers and the burst window used for metric extraction. Suggested inputs: `metrics_burst_metrics_*` plots and burst-window overlays.

SI_Fig_S4_event_histograms_placeholder.png

Figure 4: (Placeholder) Video event-time histograms and histogram-derived rate estimates supporting the burst interpretation. Suggested inputs: `compare_event_hist_*`, `compare_event_hist_rate_*`, and `compare_dn_dt_*`.

SI_Fig_S5_local_Avrami_n_placeholder.png

Figure 5: (Placeholder) Local Avrami exponent reconstructions from the video (if used), shown separately for FAPI and FAPI–TEMPO. Suggested inputs: `local_n_FAPI.png` and `local_n_FAPI_TEMPO.png`.

4 Morphology-transported kinetics bridge (static datasets)

4.1 Rationale and hierarchy of statistical support

Because only one kinetics video exists per composition (no true video replicates), statistical support for kinetics-like descriptors is obtained from the large static datasets using the replicate hierarchy:

$$\text{Grain} \rightarrow \text{ROI} \rightarrow \text{Micrograph} \rightarrow \text{Sample}.$$

The morphology-transported kinetics bridge maps static morphology distributions to an effective time axis anchored by the single-video kinetics, enabling reconstructed curves $n(t)$, dn/dt, $X(t)$, dX/dt with uncertainty bands derived from ROI/micrograph resampling.

4.2 Transport definition (overview)

In the transported representation, grain area (or equivalent radius) is used as the state variable. A monotone mapping between state and time is anchored to the video-derived kinetics ("anchor curves") so that static distributions can be expressed as time-parametrized kinetics-like curves. Different weighting schemes (count-weighted vs area-weighted, optional R^2 weighting) are evaluated for robustness.

4.3 Weighting robustness (count vs area vs optional weighting)

We compare transported curves reconstructed under:

- **count-weighted** $X(t)$ (each grain contributes equally),
- **area-weighted** $X(t)$ (each grain weighted by area),
- **optional R^2-weighted** variants when applicable (e.g., weighting by fit quality in local mappings).

The main text uses the area-weighted representation as the primary descriptor; alternative weightings and their influence on headline metrics are reported here.

4.4 Transported Avrami fits (effective descriptors)

Shifted Avrami fits are applied to the transported curves as compact effective descriptors. Because the underlying FIRA process is non-isothermal and burst-like, these parameters are interpreted comparatively (between compositions) rather than mechanistically (as literal nucleation dimensionality).

SI_Fig_S6_transport_schematic_placeholder.pdf

Figure 6: (Placeholder) Schematic of the morphology-transported kinetics bridge: video-derived anchor curves define a mapping from morphology state to effective time; static morphology distributions are transported onto the time axis to reconstruct $X(t)$ and $dX/dt(t)$ with uncertainty from ROI/micrograph resampling.

SI_Fig_S7_weighting_Xt_placeholder.png

Figure 7: (Placeholder) Weighting robustness for transported $X(t)$: count-weighted vs area-weighted vs optional weighting (if used). Suggested inputs: `SI_weighting_comparison_X_t.png`.

SI_Fig_S8_weighting_dXdt_placeholder.png

Figure 8: (Placeholder) Weighting robustness for transported $dX/dt(t)$ under count-weighted vs area-weighted vs optional weighting. Suggested inputs: `SI_weighting_comparison_dX_dt.png`.

SI_Fig_S9_transported_Avrami_fits_placeholder.png

Figure 9: (Placeholder) Transported $X(t)$ curves with continuous Avrami fits and the corresponding derivatives $dX/dt(t)$. Suggested inputs: `transported_avrami_continuous.png` and fit-parameter tables.

5 Spatially correlated fingerprints and new crowding descriptors

5.1 Canonical grain and radial averaging

For each validated grain, intragrain quantities are mapped onto normalized radius r/R (with R the equivalent grain radius) and averaged within annuli, then averaged over the ensemble. Shaded bands represent dispersion over grains (or bootstrap/ROI resampling where applicable).

5.2 Updated radial kinetic heterogeneity proxy

The main text panel (Fig. 6i) uses a radial kinetic heterogeneity proxy, computed as $\mathrm{CV}(v_{\mathrm{eff}})$ conditioned on r/R. In the updated version, the curve is plotted with a dispersion band to match the style of panels c–h. This remains a spatially correlated proxy (grain-level v_{eff} referenced to radial position), not a pixelwise transformation-rate field.

`SI_Fig_S10_radial_kinetic_QC_placeholder.png`

Figure 10: (Placeholder) QC support for the radial kinetic heterogeneity proxy: contributing grains per bin and mean ring pixels per bin. Suggested inputs: `radial_kinetic_proxy_v3_radial_kinetic_heterogeneity_qc.png` and the updated band plot.

5.3 Crowding/impingement descriptors (new panels j.i–j.iii)

Using centroids reconstructed from the crystal masks, we compute the nearest-neighbour distance (NN) and an impingement index $R_{\mathrm{eq}}/\mathrm{NN}$ and summarize both as radial profiles. NN is reported in physical units using the calibration 22.0014 px = 10 μm. These crowding descriptors quantify the external neighbourhood geometry that each grain grows into and link that geometry to kinetics via the annulus-conditioned median effective growth rate $\tilde{v}_{\mathrm{eff}}(r/R)$.

SI_Fig_S11_radial_crowding_profiles_placeholder.png

Figure 11: (Placeholder) Radial crowding profiles: median NN distance (in μm) and median impingement index versus r/R, with QC support. Suggested inputs: `radial_crowding_proxy_*_radial_nn_median_um.png`, `radial_impingement_median.png`, and `radial_crowding_qc.png`.

5.4 Caption-ready quantification of radial curves

To convert radial fingerprints into compact evidence, we quantify each curve by centre-to-mid change, mid-to-edge change, edge-minus-centre contrast, area under curve (AUC), and radii of minima. The exported SI tables provide both per-sample metrics and pairwise (FAPI minus FAPI–TEMPO) contrasts.

Table 1: (Placeholder) SI table of caption-ready quantification metrics for radial curves (per sample). Suggested input: `SI_radial_curve_metrics_per_sample.csv`.

Curve	Sample	centre→mid	mid→edge	edge−centre	AUC
(populate from exported CSV)					

Table 2: (Placeholder) SI table of pairwise radial-curve contrasts (FAPI minus FAPI–TEMPO). Suggested input: `SI_radial_curve_metrics_pairwise_FAPI_minus_TEMPO.csv`.

Curve	Δ(centre→mid)	Δ(mid→edge)	Δ(edge−centre)	ΔAUC
(populate from exported CSV)				

6 Phase-2 true local pixelwise fields (optional SI extension)

To go beyond grain-resolved proxies and move toward genuinely local spatial correlations, we construct three true local pixelwise distance fields inside each grain using distance transforms: (i) distance-to-boundary, (ii) distance-to-defect, and (iii) distance-to-nucleus. These fields are sampled against heatmap intensity, local entropy, local defect-density, and heatmap gradient magnitude to yield binned relationships for each composition. These analyses are included as SI extensions and do not alter the main text conclusions.

SI_Fig_S12_local_fields_QC_placeholder.png

Figure 12: (Placeholder) QC overlays for true local fields: distance-to-boundary, distance-to-defect, distance-to-nucleus (computed from JSON masks), shown on representative grains.

SI_Fig_S13_intensity_vs_local_fields_placeholder.png

Figure 13: (Placeholder) Binned heatmap intensity versus local distance fields (boundary/defect/nucleus), comparing FAPI and FAPI–TEMPO.

SI_Fig_S14_entropy_vs_local_fields_placeholder.png

Figure 14: (Placeholder) Binned local entropy versus local distance fields (boundary/defect/nucleus), comparing FAPI and FAPI–TEMPO.

SI_Fig_S15_defectdensity_gradient_vs_local_fields_placeholder.png

Figure 15: (Placeholder) Binned local defect density and heatmap gradient magnitude versus local distance fields (boundary/defect/nucleus), comparing FAPI and FAPI–TEMPO.

7 Outputs and reproducibility

The workflow exports:

- **Video-measured kinetics:** $X(t)$, $dX/dt(t)$, $n(t)$, $dn/dt(t)$, burst metrics, and QC plots.
- **Transported kinetics:** reconstructed curves under different weightings, main-text comparison metrics, bootstrap tables, and Avrami fit parameters.
- **Radial fingerprints:** radial descriptor profiles (panels c–i), updated radial kinetic heterogeneity proxy with dispersion, and new crowding/impingement profiles (panels j.i–j.iii), including QC support.
- **Optional local-field extensions:** distance-field maps and binned correlations against heatmap intensity/entropy/defect density/gradient magnitude.

References

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